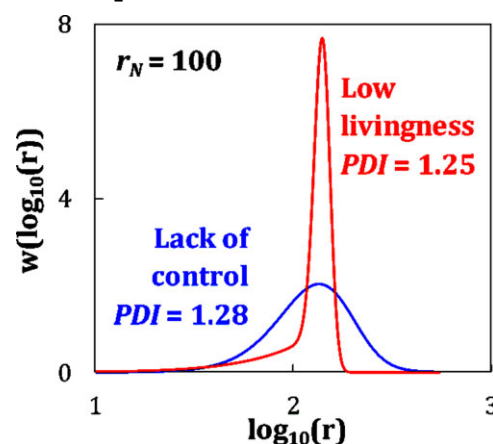


Development of Molecular Weight Distribution in ATRP with Radical Termination

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In a previous work, a model of full molecular weight distribution (MWD) for atom transfer radical polymerization with radical termination has been developed, using analogy to a series of continuous stirred tank reactors. This model assumes constant reaction rates, which is applicable to low conversion batches or steady-state continuous processes. In this work, the model is generalized to any polymerization condition, which provides variations of the reaction rates with conversion are known. Using this extended model, the effect of monomer conversion on the MWD is demonstrated, for the cases with and without radical termination. The conversion dependence of MWD is found to be significant. The broadening of distribution is correlated to the lack of control and/or the loss of livingness. Both control and livingness are necessary conditions for narrow MWD. It is also found that the effects of these two factors are very different, yielding different distribution shapes, even with similar average chain lengths (r_N) and polydispersity indexes (PDI).



1. Introduction

Controlled radical polymerization (CRP), or also termed reversible deactivation radical polymerization (RDRP) as have been recommended by IUPAC,^[1] is a type of polymerization process that allows formation of polymers with narrow chain length distribution (CLD, or equivalently molecular weight distribution, MWD). In other words, polymers produced via CRP would possess similar chain lengths to each other, as indicated by its low polydispersity index (PDI; between 1.0 and 1.5). PDI is a measure of how broadly distributed the polymer chains are, or also termed

as dispersity ($\mathcal{D}$) by IUPAC.^[2] The narrow distribution has been attributed to the reduced level of termination and to the equilibrium involved in CRP reaction mechanisms, which allows radical/active chains to be deactivated and form dormant chains. The equilibrium between the dormant and radical chains provides control to the system, where the active chain end is capped very frequently, allowing very few monomeric units to be added per activation cycle. Moreover, all the chains in CRP are initiated at the beginning of polymerization, allowing all chains to grow under the same condition, further contributing to the narrow CLD possessed.

The three most commonly studied CRP types are nitroxide-mediated polymerization (NMP),^[3] atom transfer radical polymerization (ATRP),^[4] and reversible-addition fragmentation chain transfer (RAFT).^[5] An increasing numbers of papers have been published in these areas, along with numerous reviews that summarize the progress in the area.^[6–9] The growing interest in CRP is mainly driven by its potential in industrial application to improve the

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properties of existing materials and to create new polymeric materials with carefully tailored properties. Some industrial successes have been accomplished in commercializing CRP processes, such as Kaneka's XMAP from ATRP, Rhodiblock RS of Rhodia, Arkema's Blocbuilder from NMP, etc.^[10] Researchers continue working on reaction engineering and industrial aspects of CRP processes for further commercialization. Examples are the developments of reactor technologies^[11–13] and high monomer conversion polymerization operations.^[14]

Based on the number of papers published, the most widely studied types of CRP is ATRP, followed by RAFT and NMP.^[15] For the past two decades, many different types of ATRP have been developed, such as normal ATRP, reverse ATRP, activators regenerated by electron transfer (ARGET) ATRP, initiators for continuous activator regeneration (ICAR) ATRP, etc.^[6] Despite the different types, the basic mechanisms of these ATRPs are still similar, differing only in the initiation/activation system. The basic mechanism that is applicable for all the ATRP types is shown in Figure 1, with *C* and *XC* representing the activator and deactivator catalysts, respectively. The activator and deactivator, which create an equilibrium between dormant, *PX*, and radical, *P•*, are transition metal complexes with appropriate ligand. As can be seen from Figure 1, the termination in ATRP is not completely eliminated. The suppression of termination in ATRP relies on the persistent radical effect.^[16,17]

Modeling of CRP has been of great interest because it allows deeper understanding of the reaction mechanism and kinetics, as recently demonstrated for RAFT system.^[18] There are many different modeling approaches taken in the investigation of CRP systems, from the early use of kinetic modeling and method of moments,^[19–32] to the more recent modeling studies, which utilized Monte Carlo method, PREDICI software, and other approaches.^[33–48] Out of all these studies, only some discussed the modeling of CLD, even though the knowledge of CLD is essential in determining polymer properties and in understanding reaction mechanism and kinetics. Several methods that have been employed in the literature to predict the CLD in CRP system include, but not limited to, Monte Carlo method, probability generating functions, and direct integration method.^[39–45]

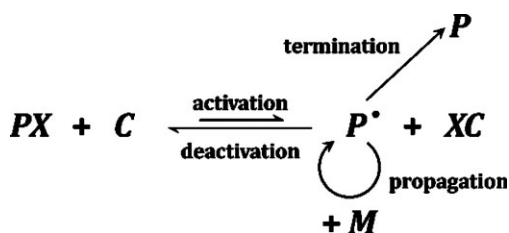


Figure 1. Basic reaction mechanism of ATRP system.

Our group has recently developed an analytical expression for the full CLD of ATRP with termination by analogously relating the activation/deactivation cycles in ATRP to a series of continuous stirred tank reactors (CSTRs).^[45] By using this analogy, the well-known concepts for CSTRs are applicable to ATRP system, allowing the derivation of full CLD, while accounting for the effect of termination, as shown in Equation (1).

$$n(r) = e^{-z}(\phi_p)^r \left[\sum_{i=0}^{\infty} \binom{r+i-1}{r} \frac{(z\phi_d)^i}{i!} + \frac{\phi_t}{\phi_d} \sum_{i=1}^{\infty} \frac{z^i}{i!} \sum_{j=1}^i \binom{r+j-1}{r} (\phi_d)^j \right] \quad (1)$$

$$z = k_a[C]t \quad (2)$$

$$\phi_p = \frac{k_p[M]}{k_p[M] + k_d[XC] + k_t[P\bullet]} \quad (3)$$

$$\phi_d = \frac{k_d[XC]}{k_p[M] + k_d[XC] + k_t[P\bullet]} \quad (4)$$

$$\phi_t = 1 - \phi_p - \phi_d \quad (5)$$

$$f_t = \frac{\phi_t}{\phi_t + \phi_d} \quad (6)$$

In that work, the average number of activation cycles that the chains have undergone, *z*, was used in lieu of reaction time, *t*, as defined by Equation (2). The different probabilities of a radical to participate in various reactions are defined by Equation (3)–(5), also illustrated in Figure 2, with the subscripts representing propagation, deactivation,

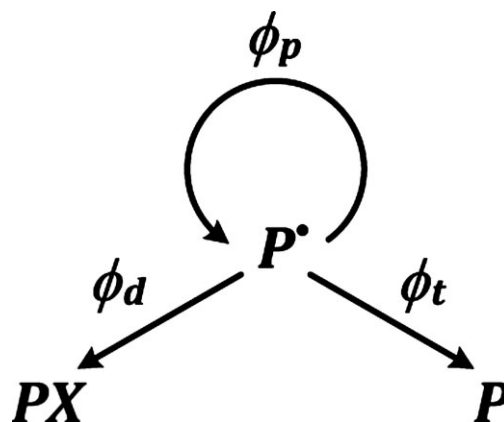


Figure 2. Possible reaction pathways of a radical to propagate to add monomeric units, to deactivate with catalyst to form dormant chain, *PX*, or to terminate with another radical to form dead chain, *P*.

and termination, respectively. The fraction of chains that are terminated in one activation cycle, f_t , is defined by Equation (6).

The analytical expression developed in that work has been utilized to study the effects of “livingness” and “control” on the broadening of CLD. However, this model was developed under the assumption that the probabilities of radical undergoing different reactions are constant. This in turn requires the reactants concentrations to be constant, which limits the applicability of Equation (1) to low conversion batch systems or steady state continuous systems. As polymerization proceeds to high conversion in batch process, significant amount of monomer is consumed and more polymer chains are irreversibly terminated to form dead chains. The depletion of monomer concentrations and accumulation of dead chains result in a significant change in reactants concentrations, thereby invalidating the use of Equation (1) for high conversion ATRP system. Furthermore, various reactions could also become diffusion controlled at high conversions, and thus the reaction rate constants could be changing too during polymerization. Variations in concentrations and rate constants change the probability parameters in the above equations. In this work, we make a great effort in extending this analytical model to account for these changes. This allows the model to be usable in predicting CLD of high conversion batch processes.

2. Model Development

The generally accepted reaction mechanism for ATRP system, shown in Figure 1, is used in the model derivation of this work. The elementary reactions are listed in Table 1. In ATRP, the key to livingness and control of polymerization lies in the equilibrium between the dormant state, PX, and the active/radical state, $P^\bullet$, which heavily favors the dormant state. The other possible side reactions not listed in Table 1 are assumed to pose negligible effect on the CLD.

For the CLD expression shown by Equation (1) to be usable for high conversion systems, the polymerization is discretized into several short intervals, during which the reactants concentrations can be assumed to be approximately constant. Therefore, the previously developed

theory could be independently applied to each interval, generating the CLD of the chains grown during that interval. These expressions are then combined to obtain the cumulative CLD for the whole period of reaction time. The discretization concept is illustrated in Figure 3.

The detailed derivation of the model is divided into two parts, first is the simplest case for the system without any termination (perfectly “living” system), while the second accounts for the effect of termination on CLD. In both cases, the intervals chosen are based on the total monomer conversion value, with 1 interval representing 1% of monomer conversion. This is done to reduce the number of CSTRs series at high conversion, since the change in concentration becomes slower towards the end of polymerization.

2.1. No Termination ($k_t = 0 \rightarrow \phi_t = 0, f_t = 0$)

For a perfectly living system under constant reactants concentrations, the full CLD of that system could be shown to follow Equation (11), with the number-average chain length and PDI expressed by Equation (12) and (13). These equations were first derived by Tobita using a random sampling technique and later by us using analogy to a series of CSTRs.^[39,45] From Equation (11), it can be seen that for the system without termination, the full CLD is described by two independent variables, namely the polymerization time, or equivalently represented by the average number of activation cycles involved, z , and the probability of radical to add monomeric unit, ϕ_p .

$$n(r) = \sum_{i=0}^{\infty} \binom{r+i-1}{r} \frac{(z(1-\phi_p))^i}{i!} \quad (11)$$

$$r_N = \frac{z\phi_p}{1-\phi_p} \quad (12)$$

$$PDI = 1 + \frac{1}{r_N} + \frac{2}{z} \quad (13)$$

With the discretization as illustrated by Figure 3, the distribution of chain growth during one interval follows Equation (11), with different values of ϕ_p and z for different intervals. For the first interval, the full CLD and the number-average chain length are shown by Equation (14) and (15), respectively.

$$n_1(r) = f(z_1, \phi_{p,1}) = \sum_{i=0}^{\infty} \binom{r+i-1}{r} \frac{(z_1(1-\phi_{p,1}))^i}{i!} \quad (14)$$

$$r_{N,1} = \frac{z_1\phi_{p,1}}{1-\phi_{p,1}} \quad (15)$$

Table 1. Reactions involved in ATRP.

Reactions in ATRP		
Activation	$P_rX + C \xrightarrow{k_a} P_r^\bullet + XC$	(7)
Deactivation	$P_r^\bullet + XC \xrightarrow{k_d} P_rX + C$	(8)
Propagation	$P_r^\bullet + M \xrightarrow{k_p} P_{r+1}^\bullet$	(9)
Termination	$P_r^\bullet + P_s^\bullet \xrightarrow{k_t} P_r + P_s$	(10)

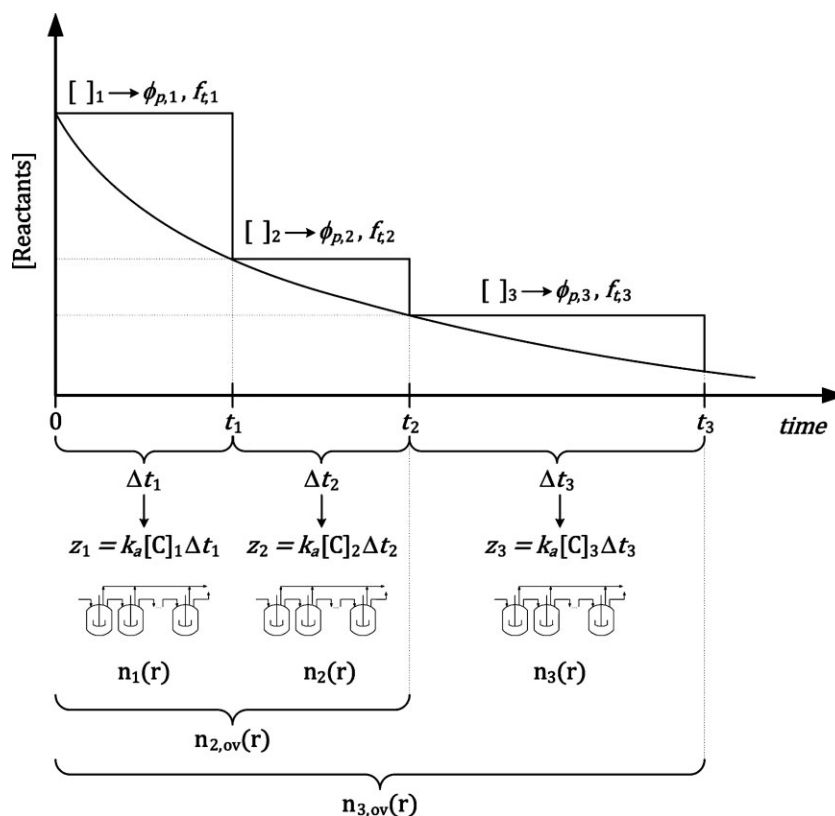


Figure 3. Discretization of polymerization period to smaller time intervals to obtain CLD.

where:

$$z_1 = k_a[C]_1 \Delta t_1 \text{ and } \phi_{p,1} = \frac{k_p[M]_1}{k_p[M]_1 + k_d[XC]_1}.$$

Similarly for the second interval, the distribution of monomer added during that interval follows Equation (16), while the number-average chain length is shown by Equation (17). The only difference between the first and second intervals is the independent variable values, z and ϕ_p . The reactants concentrations used for the calculation of z_2 and $\phi_{p,2}$ are obtained by solving a set of differential equations, shown in Appendix, at the beginning of the second interval ($t = t_1$).

$$n_2(r) = f(z_2, \phi_{p,2}) = \sum_{i=0}^{\infty} \binom{r+i-1}{r} \frac{(z_2(1-\phi_{p,2}))^i}{i!} \quad (16)$$

$$r_{N,2} = \frac{z_2 \phi_{p,2}}{1 - \phi_{p,2}} \quad (17)$$

where:

$$z_2 = k_a[C]_2 \Delta t_2 \text{ and } \phi_{p,2} = \frac{k_p[M]_2}{k_p[M]_2 + k_d[XC]_2}.$$

In order to obtain the overall distribution, the two distributions from the first and second intervals must be combined. Polymer chains that possess r monomeric units at the end of the second interval are chains that grow s units during the first interval and added $(r-s)$ units during the second interval, as illustrated by Figure 4.

In Figure 4, the dashed line is part of the polymer chain that is grown during the first interval (0 to t_1), therefore described by Equation (14). Meanwhile, the solid line represents the portion of polymer grown during the second interval (t_1 to t_2), hence follows Equation (16). The overall distribution could be obtained by taking into account all possible combinations (s from 0 to r), as mathematically shown by Equation (18). The number-average chain length at the end of the second interval could

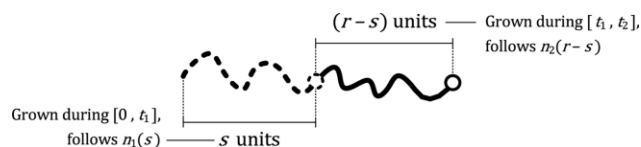


Figure 4. Chain growth history for chains with r monomeric units after two intervals in a perfectly living ATRP.

be calculated as a simple summation of number-average chain length from the first and second intervals, as shown by Equation (19). This is possible because no dead chain is considered in this case, thus both intervals involve the same chain population.

$$n_{2,ov}(r) = \sum_{s=0}^r [n_1(s) \times n_2(r-s)] \quad (18)$$

$$r_{N,2,ov} = r_{N,1} + r_{N,2} \quad (19)$$

Following the same trend, a general expression for the CLD of the chains grown during the j th interval is shown by Equation (20), resulting in the number-average chain length expression shown by Equation (21).

$$n_j(r) = f(z_j, \phi_{p,j}) = \sum_{i=0}^{\infty} \binom{r+i-1}{r} \frac{(z_j(1-\phi_{p,j}))^i}{i!} \quad (20)$$

$$r_{N,j} = \frac{z_j \phi_{p,j}}{1 - \phi_{p,j}} \quad (21)$$

where:

$$z_j = k_a[C]_j \Delta t_j \text{ and } \phi_{p,j} = \frac{k_p[M]_j}{k_p[M]_j + k_d[XC]_j}$$

$$n_{j,ov}(r) = \sum_{s=0}^r [n_{j-1,ov}(s) \times n_j(r-s)] \quad (22)$$

$$r_{N,j,ov} \equiv \sum_{r=0}^{\infty} [r \times n_{j,ov}(r)] \text{ or } r_{N,j,ov} = \sum_{i=1}^j r_{N,i} \quad (23)$$

The cumulative CLD from the beginning to the end of the j th interval (i.e. $t = [0, t_j]$) follows Equation (22). Moreover, due to the absence of dead chains in this system, the overall number-average chain length is simply the summation of those from each interval, as shown by Equation (23).

2.2. With Termination

When termination is taken into account, there are two different chain populations that have to be considered, the living (dormant) chains and the dead (terminated) chains. In our previous work, the dormant and terminated chain distributions can be separated for the system under constant reactants concentrations, as shown by Equation (24), obtained through the analogy to a series of CSTRs with by-product streams. The number-average chain length and living chain fraction (LCF) have been shown to follow Equation (25) and (26), respectively.^[45]

$$\begin{aligned} n(r) &= n^d(r) + n^t(r) \\ &= e^{-z}(\phi_p)^r \left[\sum_{i=0}^{\infty} \binom{r+i-1}{r} \frac{(z\phi_d)^i}{i!} \right. \\ &\quad \left. + \frac{\phi_t}{\phi_d} \sum_{i=1}^{\infty} \frac{z^i}{i!} \sum_{j=1}^i \binom{r+j-1}{r} (\phi_d)^j \right] \end{aligned} \quad (24)$$

where:

$$n^d(r) = f(z, \phi_p, f_t) = e^{-z}(\phi_p)^r \sum_{i=0}^{\infty} \binom{r+i-1}{r} \frac{(z\phi_d)^i}{i!}$$

$$n^t(r) = f(z, \phi_p, f_t) = e^{-z}(\phi_p)^r \frac{\phi_t}{\phi_d} \sum_{i=1}^{\infty} \frac{z^i}{i!} \sum_{j=1}^i \binom{r+j-1}{r} (\phi_d)^j$$

$$r_N = \frac{\phi_p}{f_t(1-\phi_p)} (1 - \text{LCF}) \quad (25)$$

$$\text{LCF} = \exp(-zf_t) \quad (26)$$

With the discretization, the growth of polymer chains during one interval could be expressed by the previously developed theory, as shown by Equation (24). Therefore, the distributions of dormant and dead chains grown during the first interval follow Equation (27) and (28), respectively. The number-average chain length of the total chain population and the fraction of the living chains (LCF) produced during the first interval are shown by Equation (29) and (30), respectively. Based on Equation (27) and (28), it can be observed that the independent variables affecting the distribution are the same as the case without termination, with the additional variable f_t , which is the fraction of terminated chains per cycle.

$$\begin{aligned} n_1^d(r) &= f(z_1, \phi_{p,1}, f_{t,1}) \\ &= e^{-z_1}(\phi_{p,1})^r \sum_{i=0}^{\infty} \binom{r+i-1}{r} \frac{(z_1\phi_{d,1})^i}{i!} \end{aligned} \quad (27)$$

$$\begin{aligned} n_1^t(r) &= f(z_1, \phi_{p,1}, f_{t,1}) \\ &= e^{-z_1}(\phi_{p,1})^r \frac{\phi_{t,1}}{\phi_{d,1}} \sum_{i=1}^{\infty} \frac{(z_1)^i}{i!} \sum_{j=1}^i \binom{r+j-1}{r} (\phi_{d,1})^j \end{aligned} \quad (28)$$

$$r_{N,1} = \frac{[M]_2 - [M]_1}{[PX]_1} = \frac{\phi_{p,1}}{f_{t,1}(1-\phi_{p,1})} (1 - \text{LCF}_1) \quad (29)$$

$$\text{LCF}_1 = \frac{[PX]_2}{[PX]_1} = \exp(-z_1 f_{t,1}) \quad (30)$$

Similarly, the distribution of monomer added during the second interval follows Equation (31) and (32), for the

dormant and terminated chains at the end of the second interval, respectively. As is the case without termination, the differences between chain growth in the two intervals are the concentrations used to calculate the independent variable values, z , ϕ_p , and f_t . The concentrations used to obtain the CLD of the second interval are calculated by solving the differential equations at the beginning of the second interval ($t = t_1$), and the concentrations are assumed to be constant through the second interval ($t = [t_1, t_2]$).

$$n_2^d(r) = f(z_2, \phi_{p,2}, f_{t,2}) = e^{-z_2} (\phi_{p,2})^r \sum_{i=0}^{\infty} \binom{r+i-1}{r} \frac{(z_2 \phi_{d,2})^i}{i!} \quad (31)$$

$$\begin{aligned} n_2^t(r) &= f(z_2, \phi_{p,2}, f_{t,2}) \\ &= e^{-z_2} (\phi_{p,2})^r \frac{\phi_{t,2}}{\phi_{d,2}} \sum_{i=1}^{\infty} \frac{(z_2)^i}{i!} \sum_{j=1}^i \binom{r+j-1}{r} (\phi_{d,2})^j \end{aligned} \quad (32)$$

It should be noted that the chains involved in the second interval are only those that remain as dormant chains at the end of the first interval, i.e. excluding the dead chains formed in the first interval. Therefore, the total chain populations considered in the second interval are all based on these dormant chains, $[PX]_2$. Careful consideration must be taken when defining the number-average chain length and LCF of the second interval. The number-average chain length for this interval is the average number of monomeric units added per chains that were dormant at the beginning of the second interval. Similarly, the LCF calculated in the second interval is the fraction of chains that remains as dormant chains based on these total chains. These can be mathematically expressed as Equation (33) and (34).

$$r_{N,2} = \frac{[M]_3 - [M]_2}{[PX]_2} = \frac{\phi_{p,2}}{f_{t,2}(1 - \phi_{p,2})} (1 - LCF_2) \quad (33)$$

$$LCF_2 = \frac{[PX]_3}{[PX]_2} = \exp(-z_2 f_{t,2}) \quad (34)$$

In order to combine the distribution of the two intervals, it should be noted that the dormant chains at the end of the second interval must have been dormant at the end of the first interval. However, the terminated chains could have been one of two different possibilities. First type of terminated chains consists of the chains that were dormant at the end of the first interval, participated in reactions but terminated during the second interval. Second type consists of the chains that are already terminated at the end of the first interval, hence not growing in the second interval. These chains must still be included in the overall distribution to obtain full CLD of the total populations. The three types of chains are illustrated in Figure 5. The

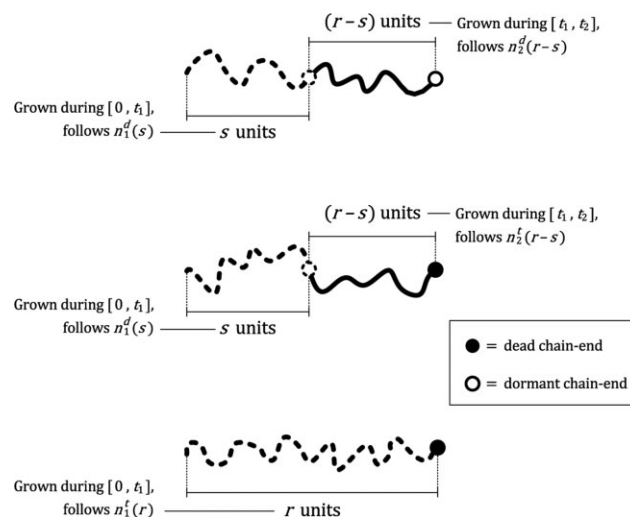


Figure 5. Different possible chain growth history for chains with r monomeric units after two intervals. Filled circle denotes dead chain-end while hollow circle denotes dormant chain-end.

portion drawn in dashed line represents the part grown during the first interval, while the solid line denotes the part grown during the second interval.

Based on Figure 5, the cumulative distributions of the dormant and dead chains at the end of the second interval could be expressed mathematically as all possible combinations of the distributions from the first and second intervals (s from 0 to r), along with the distribution of chains that were dead by the end of the first interval, shown by Equation (35) and (36). The overall distribution for the total chain populations is simply the summation of the distributions of dormant and terminated chains at the end of the second interval, as expressed by Equation (37).

$$n_{2,ov}^d(r) = \sum_{s=0}^r [n_1^d(s) \times n_2^d(r-s)] \quad (35)$$

$$n_{2,ov}^t(r) = n_1^t(r) + \sum_{s=0}^r [n_1^d(s) \times n_2^t(r-s)] \quad (36)$$

$$n_{2,ov}(r) = n_{2,ov}^t(r) + n_{2,ov}^d(r) \quad (37)$$

The overall number-average chain length could be obtained based on its definition. It could also be obtained from the combination of the two number-average chain lengths from the two intervals, while accounting the fraction of the chains that participated in the second interval, as shown by Equation (38). The overall LCF could be calculated by Equation (39) as a product of the LCFs of the first and second intervals.

$$r_{N,2,ov} \equiv \sum_{r=0}^{\infty} [r \times n_{2,ov}(r)] \quad \text{or} \quad r_{N,2,ov} = r_{N,1} + r_{N,2} LCF_1 \quad (38)$$

$$\text{LCF}_{2,\text{ov}} \equiv \sum_{r=0}^{\infty} n_{2,\text{ov}}^d(r) \quad \text{or} \quad \text{LCF}_{2,\text{ov}} = \frac{[\text{PX}]_3}{[\text{PX}]_1} = \text{LCF}_1 \times \text{LCF}_2 \quad (39)$$

The expression for the third interval could be obtained by following the same approach. The full CLD expressions for chain grown during the third interval for dormant and terminated chains are shown by Equation (40) and (41), respectively. The overall distributions for the dormant and dead chains are obtained by utilizing the cumulative distribution up to the second interval, shown by Equation (42) and (43). The total number-average chain length and LCF at the end of the third interval could be estimated based on Equation (44) and (45), respectively.

$$\begin{aligned} n_3^d(r) &= f(z_3, \phi_{p,3}, f_{t,3}) \\ &= e^{-z_3} (\phi_{p,3})^r \sum_{i=0}^{\infty} \binom{r+i-1}{r} \frac{(z_3 \phi_{d,3})^i}{i!} \end{aligned} \quad (40)$$

$$\begin{aligned} n_3^t(r) &= f(z_3, \phi_{p,3}, f_{t,3}) \\ &= e^{-z_3} (\phi_{p,3})^r \frac{\phi_{t,3}}{\phi_{d,3}} \sum_{i=1}^{\infty} \frac{(z_3)^i}{i!} \sum_{j=1}^i \binom{r+j-1}{r} (\phi_{d,3})^j \end{aligned} \quad (41)$$

$$n_{3,\text{ov}}^d(r) = \sum_{s=0}^r [n_{2,\text{ov}}^d(s) \times n_3^d(r-s)] \quad (42)$$

$$n_{3,\text{ov}}^t(r) = n_{2,\text{ov}}^t(r) + \sum_{s=0}^r [n_{2,\text{ov}}^d(s) \times n_3^t(r-s)] \quad (43)$$

$$r_{N,3,\text{ov}} \equiv \sum_{r=0}^{\infty} [r \times n_{3,\text{ov}}(r)] \quad \text{or} \quad r_{N,3,\text{ov}} = r_{N,2,\text{ov}} + r_{N,3} \text{LCF}_{2,\text{ov}} \quad (44)$$

$$\text{LCF}_{3,\text{ov}} \equiv \sum_{r=0}^{\infty} n_{3,\text{ov}}^d(r) \quad \text{or} \quad \text{LCF}_{3,\text{ov}} = \text{LCF}_{2,\text{ov}} \times \text{LCF}_3 \quad (45)$$

Following the same trend, general expressions for the overall distributions of dormant and dead chains after j th interval could be shown to follow Equation (46) and (47), with the distribution of the total chain populations shown by Equation (48). The total number-average chain length and LCF at the end of the j th interval could be expressed as Equation (49) and (50).

$$n_{j,\text{ov}}^d(r) = \sum_{s=0}^r [n_{(j-1),\text{ov}}^d(s) \times n_j^d(r-s)] \quad (46)$$

$$n_{j,\text{ov}}^t(r) = n_{(j-1),\text{ov}}^t(r) + \sum_{s=0}^r [n_{(j-1),\text{ov}}^d(s) \times n_j^t(r-s)] \quad (47)$$

$$n_{j,\text{ov}}(r) = n_{j,\text{ov}}^d(r) + n_{j,\text{ov}}^t(r) \quad (48)$$

$$r_{N,j,\text{ov}} \equiv \sum_{r=0}^{\infty} [r \times n_{j,\text{ov}}(r)] \quad \text{or} \quad r_{N,j,\text{ov}} = \sum_{i=1}^j [r_{N,i} \times \text{LCF}_{(i-1),\text{ov}}] \quad (49)$$

$$\text{LCF}_{j,\text{ov}} \equiv \sum_{r=0}^{\infty} n_{j,\text{ov}}^d(r) \quad \text{or} \quad \text{LCF}_{j,\text{ov}} = \prod_{i=1}^j \text{LCF}_i \quad (50)$$

For both cases without and with termination, the overall distribution of polymer chains, represented by Equation (22) and (48), could be obtained once the full CLD of each interval is obtained. By using the approach outlined in this section, these effects could be studied for high conversion batch systems, in which the reactants concentrations are changing, resulting in variation of reaction probability values. The resulting average properties obtained from this method are compared with those obtained by method of moments to ensure the discretization do not introduce significant errors. Method of moments has been relied on to study the kinetics of various CRP systems as discussed in the Introduction section. The moments equations are included in the Appendix.

Similar to our previous work, the control level in a polymerization system is quantified by how many monomers are added to chain per activation cycle ($=\phi_p/(1-\phi_p)$). Higher growth probability, ϕ_p , value implies more monomer is added per activation cycle, which in turn means a low level of control. The livingness of a system is quantified by the amount of chains that are still living, or the LCF, which depends on both the number of activation cycles, z , and the fraction of terminated chains per cycle, f_t .

It should be noted that the model developed here is a general model for all types of ATRP system (e.g. normal ATRP,ARGET ATRP, ICAR ATRP, etc.). The only information that the model requires is the dependence of concentrations and reaction rate constants with respect to monomer conversion, or more generally, the reaction rates as functions of conversion. The reaction rate constants are known to change as the viscosity of the system increases due to diffusion limitations, as have been noted by various researchers.^[30–32,47,48] In this work, for purpose of simplicity, only changes in concentrations as functions of conversion are considered, while the reaction rate constants are considered to be unaffected.

3. Simulation Results and Discussion

The reaction rate constants used in all simulations conducted in this work are listed in Table 2. These reaction rate constants are the typical values for ATRP of MMA at 60 °C.

The CLDs predicted by analytical model, which assumes constant reaction conditions, is compared to that predicted by the model developed in this work, which accounts for the

Table 2. Reaction rate constants used in simulation of ATRP system.

Reaction rate constants	Values
k_p (L mol ⁻¹ s ⁻¹) ^[49]	834
k_a (L mol ⁻¹ s ⁻¹)	1.0
k_d (L mol ⁻¹ s ⁻¹)	10 ⁶
k_t (L mol ⁻¹ s ⁻¹)	0 or 10 ⁸

change in reaction probabilities (ϕ_p and f_t) as functions of conversion. The comparison is done for both cases with and without termination, in order to investigate the effect of conversion on the development of CLD in batch processes, where the change in reactants concentrations could not be neglected. In most cases, for the conversion-dependent model, or also referred to as the extended model, the reaction probability values are recalculated every 1% conversion by solving the differential equations derived from reaction kinetics.

3.1. Without Termination

For the cases without termination, there are two independent variables affecting the distribution, namely the number of activation cycles (z) and the probability of growth (ϕ_p). The importance of reaction probability values in predicting CLD for a perfectly living system is investigated by comparing the analytical model with constant reaction probabilities with the solution of the extended model with variable reaction probabilities. The comparison is shown in Figure 6a, with the simulation results tabulated in Table 3. The discretization of the simulation parameter, ϕ_p , for the living system is illustrated by a step function in Figure 6b, while the dashed line represents the evolution of ϕ_p calculated from the reactants concentrations obtained by solving the set of differential equations.

Table 3. Simulation results for perfectly living ATRP corresponding to Figure 6a.

$\phi_p = 0.63-0.02$	Polydispersity	
	Analytical	Extended
$r_N = 100$, $x = 50\%$	1.04	1.03
$r_N = 200$, $x = 100\%$	1.02	1.01

For the analytical model, only a constant value of reaction probability can be used per simulation, which is chosen to be the value calculated at the beginning of the polymerization ($x = 0\%$) for this initial comparison.

As can be observed from Figure 6a above, the difference between the CLDs obtained from analytical and extended models becomes more obvious at higher conversion. This is because the concentrations deviate further from their initial values as conversion increases. The analytical model generates a slightly broader distribution than the extended one because as the monomer is consumed, the propagation probability, ϕ_p , decreases, as shown in Figure 6b. In terms of control level, this implies a more controlled condition because lower ϕ_p means fewer monomeric units is added per activation cycle ($=\phi_p/(1-\phi_p)$).

Beside the initial value of ϕ_p , two other ϕ_p values are tested for the analytical model in comparison with the extended model. These values are calculated based on the reactants concentrations at 50% and 100% conversions for the system with $r_N = 200$ (corresponding to 100% conversion). The comparison is shown in Figure 7a. The evolution of probability values with monomer conversion is shown in Figure 7b.

From Figure 7a, the use of ϕ_p calculated at halfway conversion ($x = 50\%$) for analytical model approximates the CLD obtained by the extended model. However, this is only due to the relatively simple relationship between ϕ_p

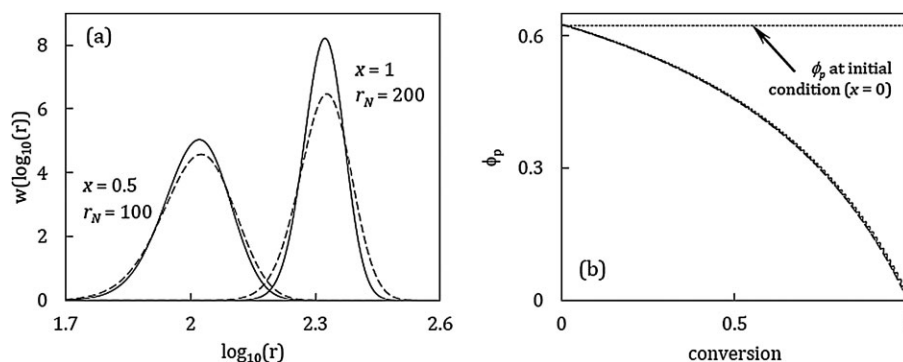


Figure 6. Simulation results of a perfectly living ATRP system. $[M] = 4.7$ M, $[M]:[RX]:[C]:[XC] = 200:1:1:0.1$. a) Comparison of CLDs obtained by analytical and extended models after 50 and 100% conversion. Solid lines denote extended model with variable probability, dashed lines denote analytical model with probability calculated at $x = 0\%$. b) Evolution of propagation probability with conversion.

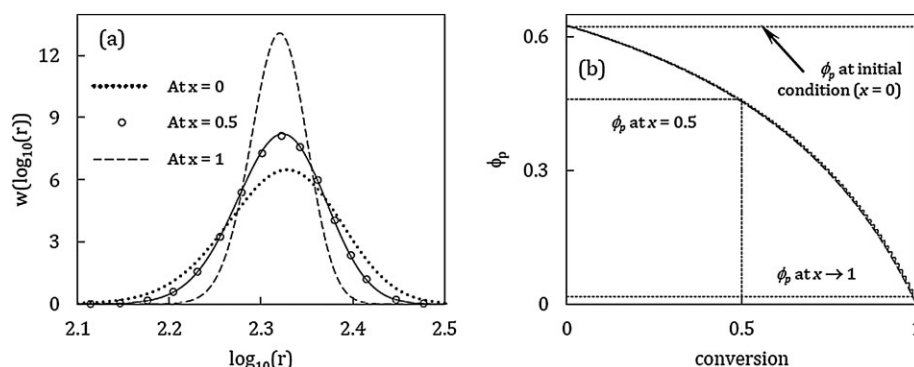


Figure 7. a) Simulation results of CLD for perfectly living ATRP system after 100% conversion ($r_N = 200$). Solid line denotes extended model with variable probability, dashed lines denote analytical model with constant probability calculated from various conversions. b) Evolution of propagation probability with conversion. Polymerization recipe: $[M] = 4.7\text{ M}$, $[M]:[RX]:[C]:[XC] = 200:1:1:0.1$.

and conversion for a perfectly living system, since the deactivator concentration does not change significantly with conversion in the absence of termination. Therefore, the propagation probability for a perfectly living system could be simplified from Equation (3) to follow Equation (51) as a function of conversion. This expression results in a linear function of level of control ($=\phi_p/(1-\phi_p)$) with respect to conversion, which explains why the simulation parameters in the analytical model calculated at the half-way conversion point gives a good estimate for the CLD predicted by the extended model.

$$\begin{aligned}\phi_p = f(x) &= \frac{k_p[M]}{k_p[M] + k_d[XC] + k_t[P^*]} \xrightarrow{\text{No termination, } k_t=0} \frac{\phi_p}{1-\phi_p} \\ &= \frac{k_p[M]_0}{k_d[XC]_0} (1-x)\end{aligned}\quad (51)$$

When termination is taken into account, the concentrations of other reactants also change with conversion, resulting in a more complicated non-linear dependence of the simulation parameters on conversion. Therefore, the analytical model with parameters calculated at any conversion point would not be expected to exhibit similar CLDs as the extended model. For simplicity purpose, all subsequent comparisons of analytical and extended models will only be shown for the analytical model calculated with parameters based on the initial polymerization condition.

3.2. With Termination

When termination is considered, there are three independent variables that are required to obtain the CLD of ATRP system by using Equation (24), namely the average number of activation cycles (z), growth probability in a cycle (ϕ_p), and fraction of terminated chains in a cycle (f_t). Conversion, or equivalently number-average chain length, is set to

be an independent variable for comparison purpose to replace time, and equivalently z . Similar to the case without termination, the CLD predictions from analytical and extended models are compared by using initial simulation parameters for the analytical model, as shown in Figure 8, with the simulation results summarized in Table 4. The dependence of the simulation parameters, ϕ_p and f_t , with conversion are shown in Figure 9, with the discretization of these parameters for the extended model shown as step functions.

The comparison presented in Figure 8 shows that the difference in CLDs from the extended and analytical models is more obvious at higher conversion. However, in this case the difference is not only caused by the monomer consumption but also by the accumulation of dead chains in the system. Based on Table 4, the amount of living chains predicted by the analytical model is much

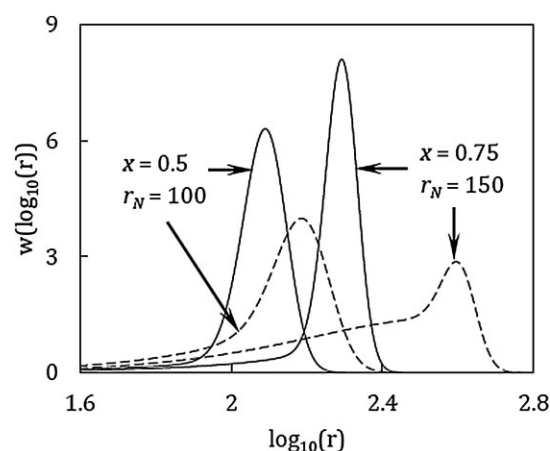


Figure 8. Comparison of CLDs obtained by analytical and extended models after 50% and 75% conversion. Recipe: $[M] = 4.7\text{ M}$, $[M]:[RX]:[C]:[XC] = 200:1:1:0.1$. Solid lines denote conversion-dependent solution of the extended model, dashed lines denote analytical model with parameters calculated at initial condition.

Table 4. Simulation results for ATRP system by model with termination, corresponding to Figure 8.

$\phi_p = 0.62-0.02$ $f_t = 1E-2 - 5E-5$	Polydispersity		LCF	
	Analytical	Extended	Analytical	Extended
$r_N = 100, x = 50\%$	1.33	1.16	40%	73%
$r_N = 150, x = 75\%$	1.69	1.20	9%	64%

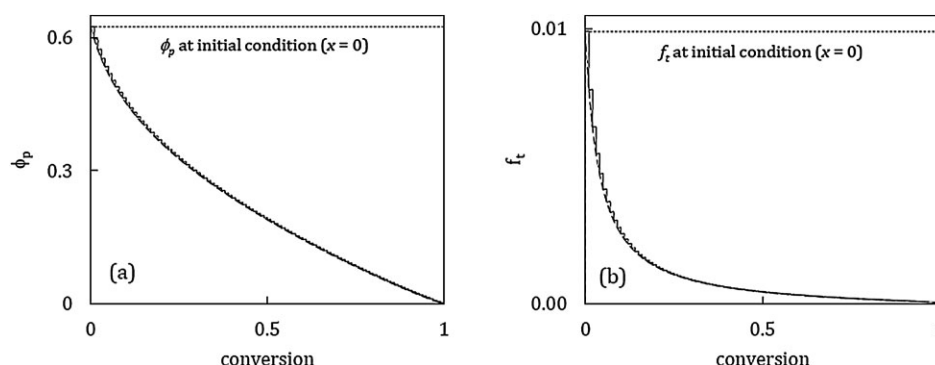


Figure 9. Evolution of simulation parameters with conversion, corresponding to Figure 8. a) Propagation probability. b) Fraction of dead chains per activation cycle.

less than that predicted by the extended model. This is also evident from the longer, more obvious tails on the CLDs predicted by the analytical model (dashed lines) shown in Figure 8, indicating large amount of dead chains that stopped growing prematurely. The overestimation of dead chains amount in the analytical model is due to its use of the initial f_t and thus not taking account the rapid decrease of f_t with conversion, as shown by Figure 9b. Therefore, for analytical model with parameters calculated based on the initial condition, termination is assumed to occur at its maximum rate, yielding lower livingness than if the conversion dependence of the dead chains formation rate is considered.

It should be noted that while ϕ_p is a good indicator of how controlled the system is, the simulation parameter f_t is not a good measure of the livingness in the system. The level of livingness can be better gauged by the fraction of living chains remaining at that conversion (LCF). Moreover, higher degree of control comes at the cost of slower polymerization rate, since fewer monomeric units are added per cycle in a more controlled system.

3.3. Lack of Control Versus Loss of Livingness

In our previous study, the broadening of CLD is attributed to the low level of control or low livingness under constant reactants concentrations.^[45] In this section, the degree of

control and livingness are varied by varying the amount of catalyst present in the system to observe the effect on the CLD with the parameters changing with conversion. The ratio between catalyst and deactivator is kept constant between the two cases at 10:1 ratio. The resulting CLD and the average properties are reported in Figure 10 and Table 5, respectively. The dependence of simulation

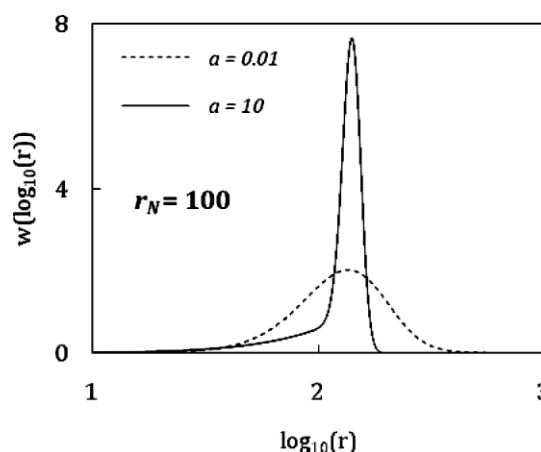
Figure 10. CLD of polymers simulated with the extended model, accounting for conversion dependence, at 50% conversion ($r_N = 100$), with different catalyst and deactivator concentrations. Recipe: $[M] = 4.7 \text{ M}$, $[M]:[RX]:[C]:[XC] = 200:1:a:0.1a$.

Table 5. Simulation results for ATRP systems corresponding to Figure 10.

$x = 50\%$	Polydispersity	LCF
$a = 0.01$	1.28	98%
$a = 10$	1.25	52%

parameters, ϕ_p and f_t , with conversion are shown in Figure 11.

It should be noted that despite the similar value of PDIs between the two cases presented in Table 5, 1.25 and 1.28, the distributions of polymer chains from these two cases are very different. The broadening observed in Figure 10 for the system with very low catalyst and deactivator concentrations (dashed line, $a = 0.01$) could be attributed to the non-uniformity in chain growth, as indicated by the high value of growth probability, ϕ_p , shown in Figure 11. The high value of ϕ_p implies that a large number of monomeric units are added in one activation cycle, thus the chains do not grow at a uniform rate. This translates to a poorly controlled system, leading to broadening of the CLD observed.

On the other hand, for the case with high catalyst and deactivator concentrations (solid line, $a = 10$), the system shows a good level of control, indicated by the narrow distribution shown in Figure 10, and by the low value of ϕ_p in Figure 11. Contrary to the previous case, the low value of ϕ_p indicates only a few monomeric units are added to a chain in one activation cycle, leading to a well-controlled system giving a narrow distribution. However, the system involves a large degree of termination, as indicated by the long tail in Figure 10 and low LCF in Table 5.

Upon observation, comparison between the amount of living chains and the value of f_t of the two cases seem to contradict each other. A system with low catalyst and

deactivator concentrations has higher f_t value, yet exhibits high degree of livingness (higher LCF). It should be noted that the degree of livingness, quantified by LCF, depends on both f_t and the average number of activation cycles (z) required to achieve a certain conversion or average chain length. In a well-controlled system, the average number of activation cycles required is higher because there are fewer monomeric units added per activation cycles (low ϕ_p). This shows that there exists a relation between the level of control and livingness. When termination is considered, very high degree of control (high number of activation cycles) could be detrimental to livingness of the system.

From the comparison, it becomes evident that the broadening of CLD could be attributed to the poor control, such as exhibited by the case with low catalyst and deactivator concentrations in Figure 10, or it could be due to the loss of livingness, observed in the case with high catalyst and deactivator concentrations in Figure 10. The effects of livingness and control on the CLD are not independent from each other, since a high degree of control could lead to the loss of livingness, resulting in broadening of the distribution. Moreover, the knowledge of average properties itself might not be enough, since two polymer populations with similar average chain length and PDI could have different distributions, as clearly shown in Figure 10, depending on what causes the broadening.

4. Conclusion

Growing research interests in ATRP for the past two decades is driven by its ability to polymerize chains with similar lengths, creating a narrowly distributed polymer chains. The knowledge of CLD (or equivalently MWD) is crucial in determining polymer properties and is

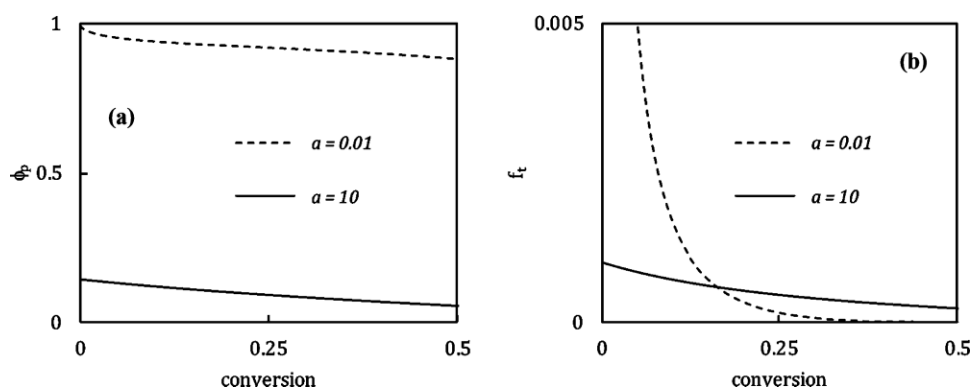


Figure 11. Evolution of simulation parameters with conversion corresponding to Figure 10. a) Propagation probability. b) Fraction of dead chains per activation cycle.

potentially beneficial in gaining better understanding of the reaction mechanism. In this work, a previously developed analytical model for CLD in ATRP is extended by applying the expression to small time intervals, during which the concentrations could be approximated as constant values. The cumulative CLD is obtained through combination of all the analytically calculated CLDs throughout the polymerization. By doing so, the variations of reactants concentrations with conversion are accounted for in this approach.

Through simulation using this extended model, it has been shown that the broadening of CLD in ATRP could be attributed to the lack of control in the system or the loss of livingness during the polymerization. Although the two phenomena result in the increase of polydispersity, the shape of the distribution is not the same. The lack of control results in a broader distribution due to non-uniform growth, while the system with low livingness still exhibits a very narrow distribution for the dormant chains, along with extended tails due to the presence of dead chains that stopped growing prematurely. Moreover, it has been shown that the control and livingness are not independent from each other. High degree of control could not only result in long polymerization time, but also could lead to loss of livingness. Control and livingness are both necessary conditions for a narrow CLD in ATRP, as well as in other type of CRP processes.

This approach, which yields a conversion-dependent model, provides a general model to predict the CLD of any type of ATRP, e.g. normal ATRP, ARGET ATRP, ICAR ATRP, etc. as long as the change in reactants concentrations as functions of monomer conversion are known. In other words, by obtaining the reaction conditions as functions of monomer conversion through any approach, the CLD of that system could be predicted using this extended model, regardless of how the activator/deactivator equilibrium is achieved. Moreover, changes in reaction rate constants due to diffusion-limitation could also be incorporated in the extended model to account for gel-effect and glass effect that could occur in any radical polymerization processes.

Appendix

The differential equations used in the recalculation of simulation parameters in this work were derived from reactions listed in Table 1. These differential equations are obtained from the mass balance of each chemical species. The mass balance for monomer, which could also be rewritten for conversion, is shown in Equation (A1). The mass balances for activator and deactivator catalyst, [C] and [XC], and for dormant [PX], radical [P•], and dead [P] chain overall populations are shown in Equation (A2)–(A6), respectively.

The simulation parameters were calculated by using the concentration profiles obtained from simultaneously solving these differential equations.

$$\frac{d[M]}{dt} = -k_p[M][P^\bullet] \rightarrow \frac{dx}{dt} = k_p[P^\bullet](1-x) \quad (A1)$$

$$\frac{d[C]}{dt} = k_d[P^\bullet][XC] - k_a[PX][C] \quad (A2)$$

$$\frac{d[XC]}{dt} = k_a[PX][C] - k_d[P^\bullet][XC] \quad (A3)$$

$$\frac{d[PX]}{dt} = k_d[P^\bullet][XC] - k_a[PX][C] \quad (A4)$$

$$\frac{d[P^\bullet]}{dt} = k_a[PX][C] - k_d[P^\bullet][XC] - k_t[P^\bullet][P^\bullet] \quad (A5)$$

$$\frac{d[P]}{dt} = k_t[P^\bullet][P^\bullet] \quad (A6)$$

The use of method of moments in determining the average properties of polymers produced by ATRP has been thoroughly explained in previous work.^[20] Therefore, only brief discussion is offered in this work. The definition of *j*th order moments for dormant, radical, and dead polymer chains are shown by Equation (A7)–(A9).

$$[PO_jX] = \sum_{r=0}^{\infty} r^j [PX_r] \quad (A7)$$

$$[PO_j^\bullet] = \sum_{r=0}^{\infty} r^j [P_r^\bullet] \quad (A8)$$

$$[PO_j] = \sum_{r=0}^{\infty} r^j [P_r] \quad (A9)$$

The mass balances of the three chain types, dormant, radical, and dead, are written for chains with *r*-monomeric units, shown in Equation (A10)–(A12), respectively.

$$\frac{d[PX_r]}{dt} = k_d[P_r^\bullet][XC] - k_a[PX_r][C] \quad (A10)$$

$$\begin{aligned} \frac{d[P_r^\bullet]}{dt} = & k_p[M]([P_{r-1}^\bullet] - [P_r^\bullet]) + k_a[PX_r][C] \\ & - k_d[P_r^\bullet][XC] - k_t[P_r^\bullet][P_r^\bullet] \end{aligned} \quad (A11)$$

$$\frac{d[P_r]}{dt} = k_t[P_r^\bullet][P_r^\bullet] \quad (A12)$$

The differential equations for zeroth, first, and second moments for the three chain populations could be obtained

by combining the mass balances in Equation (A10)–(A12) with the moments' definitions from Equation (A7)–(A9). These differential equations are shown by Equation (A13)–(A21). It should be noted that the zeroth moments yields the overall concentrations of these chains, i.e. Equation (A13)–(A15) are the same as Equation (A4)–(A6).

$$\frac{d[\text{PQ}_0\text{X}]}{dt} = \frac{d[\text{PX}]}{dt} = k_d[\text{PQ}_0^\bullet][\text{XC}] - k_a[\text{PQ}_0\text{X}][\text{C}] \quad (\text{A13})$$

$$\frac{d[\text{PQ}_0^\bullet]}{dt} = \frac{d[\text{P}^\bullet]}{dt} = k_a[\text{PQ}_0\text{X}][\text{C}] - k_d[\text{PQ}_0^\bullet][\text{XC}] - k_t[\text{PQ}_0^\bullet][\text{PQ}_0^\bullet] \quad (\text{A14})$$

$$\frac{d[\text{PQ}_0]}{dt} = \frac{d[\text{P}]}{dt} = k_t[\text{PQ}_0^\bullet][\text{PQ}_0^\bullet] \quad (\text{A15})$$

$$\frac{d[\text{PQ}_1\text{X}]}{dt} = k_d[\text{PQ}_1^\bullet][\text{XC}] - k_a[\text{PQ}_1\text{X}][\text{C}] \quad (\text{A16})$$

$$\frac{d[\text{PQ}_1^\bullet]}{dt} = k_p[\text{M}][\text{PQ}_0^\bullet] + k_a[\text{PQ}_1\text{X}][\text{C}] - k_d[\text{PQ}_1^\bullet][\text{XC}] - k_t[\text{PQ}_1^\bullet][\text{PQ}_1^\bullet] \quad (\text{A17})$$

$$\frac{d[\text{PQ}_1]}{dt} = k_t[\text{PQ}_0^\bullet][\text{PQ}_1^\bullet] \quad (\text{A18})$$

$$\frac{d[\text{PQ}_2\text{X}]}{dt} = k_d[\text{PQ}_2^\bullet][\text{XC}] - k_a[\text{PQ}_2\text{X}][\text{C}] \quad (\text{A19})$$

$$\frac{d[\text{PQ}_2^\bullet]}{dt} = k_p[\text{M}]([\text{PQ}_0^\bullet] + 2[\text{PQ}_1^\bullet]) + k_a[\text{PQ}_2\text{X}][\text{C}] - k_d[\text{PQ}_2^\bullet][\text{XC}] - k_t[\text{PQ}_2^\bullet][\text{PQ}_2^\bullet] \quad (\text{A20})$$

$$\frac{d[\text{PQ}_2]}{dt} = k_t[\text{PQ}_0^\bullet][\text{PQ}_2^\bullet] \quad (\text{A21})$$

By solving Equation (A13)–(A21) simultaneously, the evolution of zeroth, first, and second order moments with time could be obtained. From these moments, the number-average chain length, weight-average chain length, and polydispersity could be calculated by using Equation (A22), (A23), and (A24), respectively.

$$r_N = \frac{[\text{PQ}_1\text{X}] + [\text{PQ}_1^\bullet] + [\text{PQ}_1]}{[\text{PQ}_0\text{X}] + [\text{PQ}_0^\bullet] + [\text{PQ}_0]} \quad (\text{A22})$$

$$r_W = \frac{[\text{PQ}_2\text{X}] + [\text{PQ}_2^\bullet] + [\text{PQ}_2]}{[\text{PQ}_1\text{X}] + [\text{PQ}_1^\bullet] + [\text{PQ}_1]} \quad (\text{A23})$$

$$\text{PDI} = \frac{r_W}{r_N} \quad (\text{A24})$$

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